

# Irrigation Management using Machine Learning Regressors of Aggregated Soil Moisture Data from a Single Growing Season.



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# Introduction

There are many benefits for monitoring soil moisture in irrigated agriculture:

1. Energy savings
2. Water savings
3. Better management of crop health, quality and yield
4. Fertilizer savings
5. Less wear and tear on irrigation system
6. Less labor
7. Many environmental benefits



# Current Situation

In practice, soil moisture is often difficult to interpret for irrigators to provide actionable decisions about irrigation

Challenges include:

- Long underground irrigation lines make it impossible to know when water reaches the crop and difficult to see problems
- Hilly terrain and slopes have micro-climates
- Variability in soil types, soil water distribution and hydrological characteristics make it difficult to to interpret data spatially
- In the US, water rights and water allocation (drought) sometimes dictate when and how much water an irrigator can use



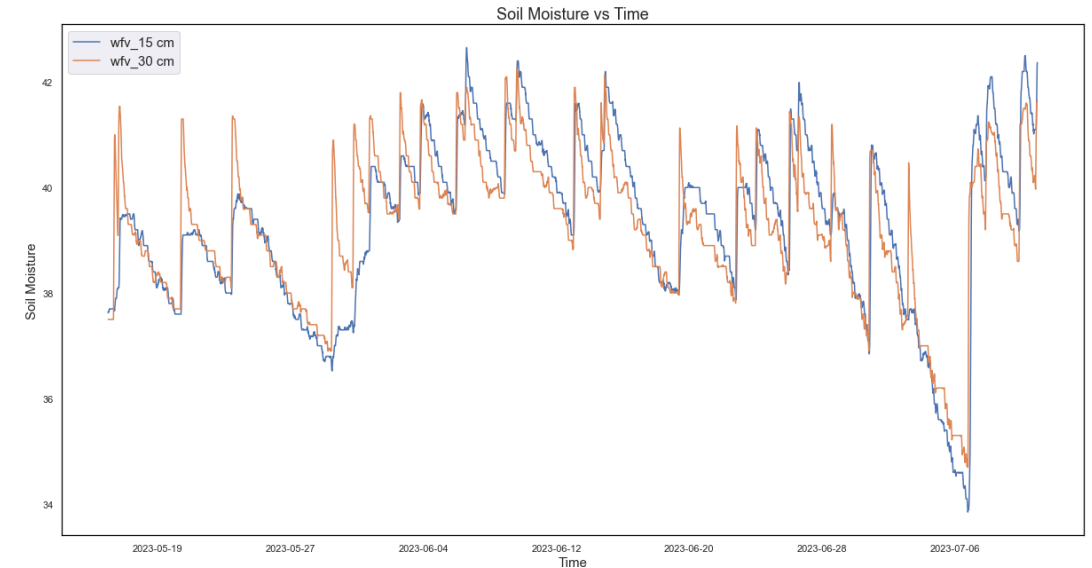
# Research Goal

The goal of this work is to provide a model that answers a simple actionable question.

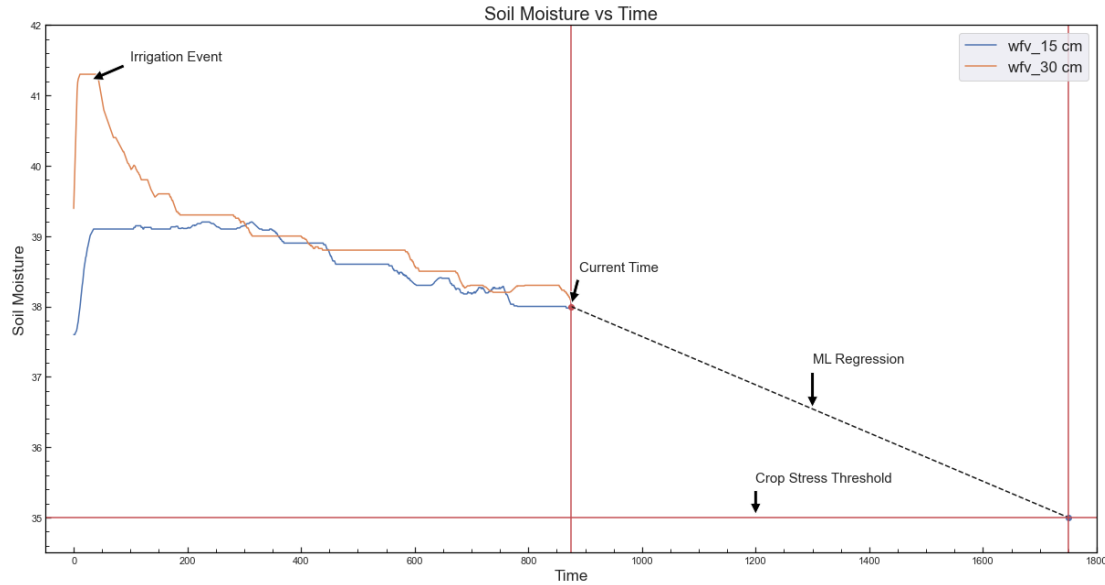
**“If the crop does not receive any water from irrigation or rain, when will it become stressed?”**

As is typical of most farming sites, by choice only a limited amount of information is available:

- Time series soil moisture at two depths, 15 and 30 cm.
- Evapotranspiration and precipitation time series data
- Regional soil survey information



# Model Overview



**Can we use machine learning regressors to predict future soil moisture values?**

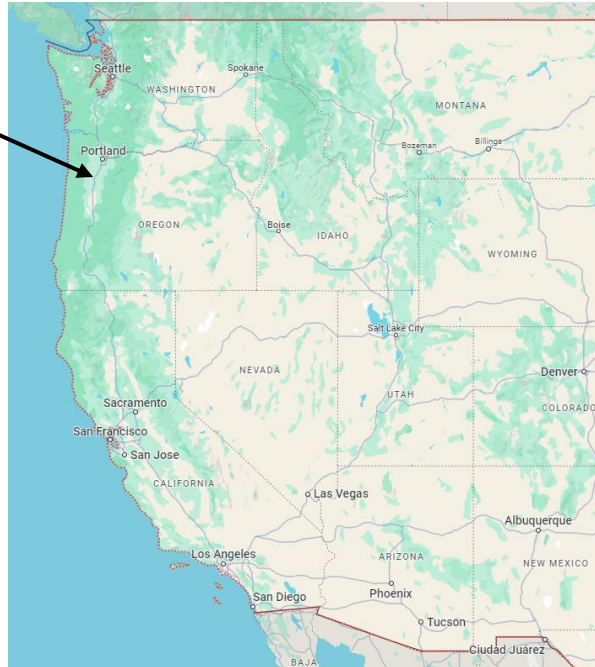
- The model only looks at the dewatering portion of the time series soil moisture data.
- The movement of the soil water is transient flow

The training data set contains:

- 1) Pairs of measured soil moistures separated by a time interval,  $t_s$ .
- 2) The sum of the hourly ET between the pair of soil moisture values.
- 3) Hydraulic gradients calculated from soil moisture and soil survey information

# Test Site

Wilsonville,  
Oregon



- Location: Wilsonville, Oregon
- Crop: 5-year-old blueberries
- Time period: May 15 – Sept. 15, 2023
- Soil: Well drained silt loam  
Willamette Valley flood  
sediment
- Irrigation: Sub-surface drip



# Sensor and Data Acquisition



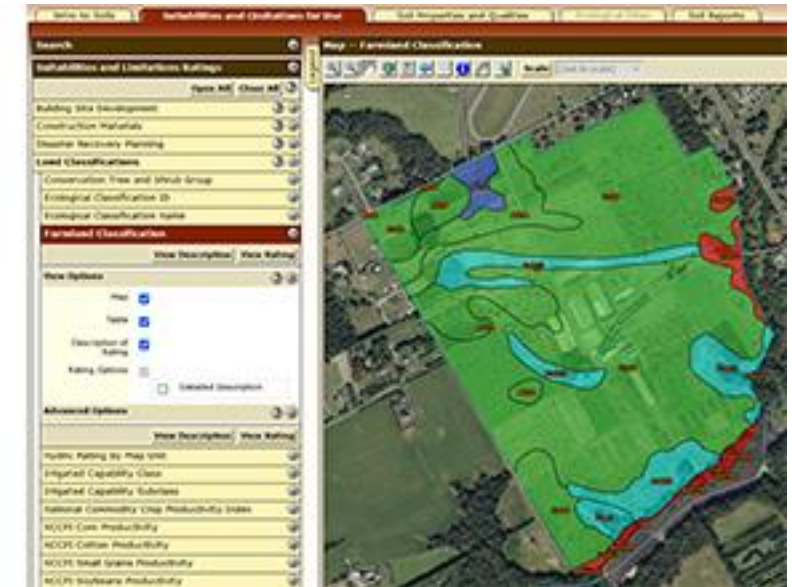
- Two Stevens HydraProbe soil sensors were placed at 15 and 30 cm deep in the root ball between two subsurface emitters
- The sensors were connected to a Steelhead data logger collecting data every 5 minutes.



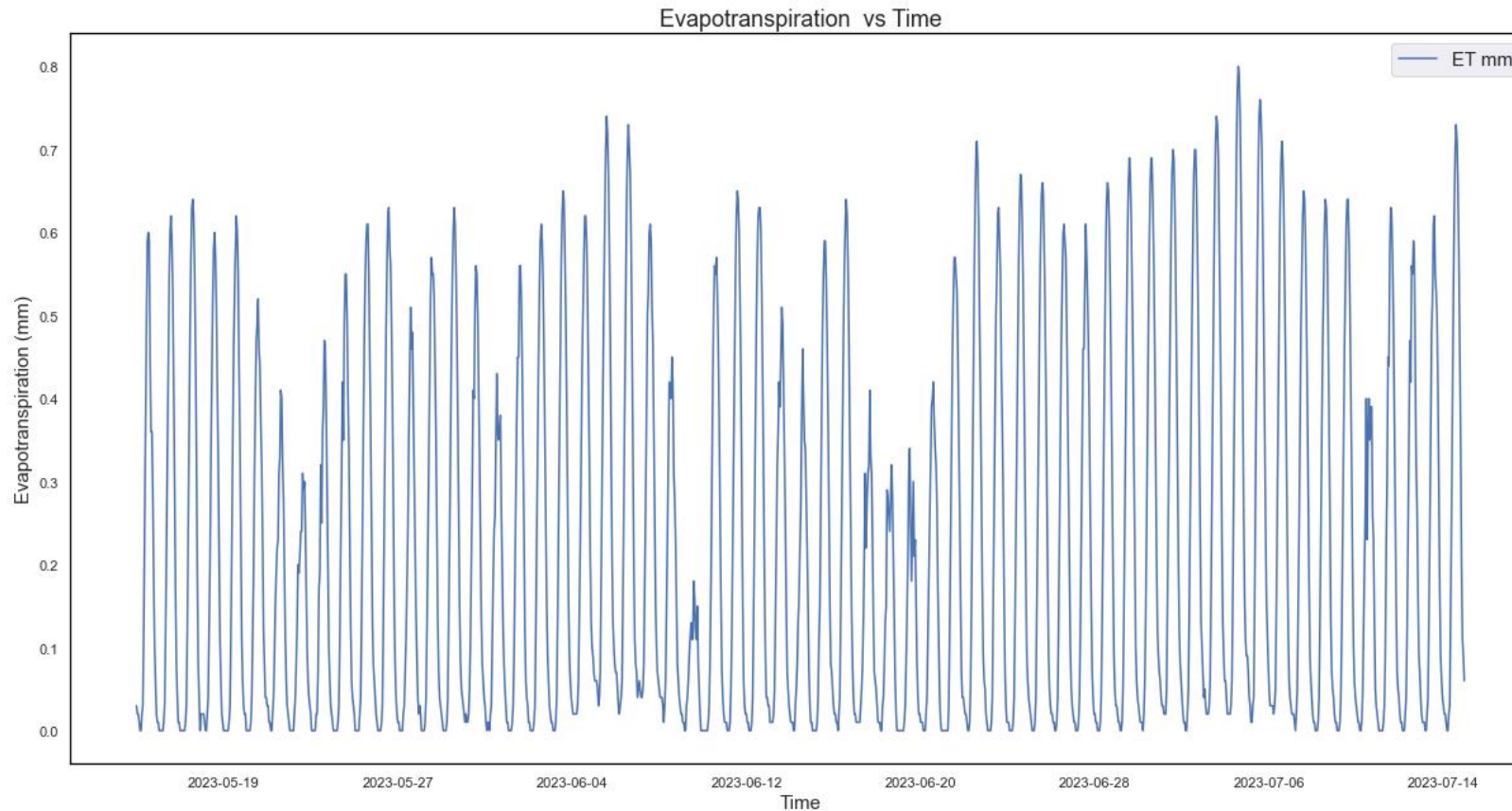
# ET and Soil Properties (Open Source Data)

- The soil water retention parameters were provided by the United States Department of Agriculture Web Soil Survey for Clackamas County Oregon

| Parameters | $\theta_s$ (wfv) | $\theta_r$ (wfv) | $\alpha$ | n    | $K_s$ (cm/s) | $l\ell$ |
|------------|------------------|------------------|----------|------|--------------|---------|
| Silt Loam  | 0.439            | 0.065            | 0.005058 | 1.66 | 1.75         | 2.317   |

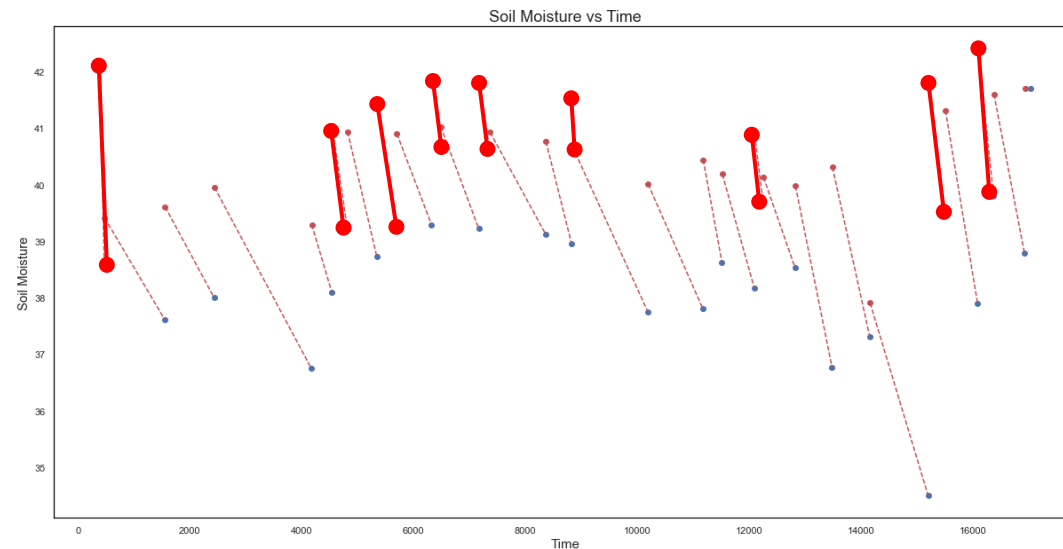
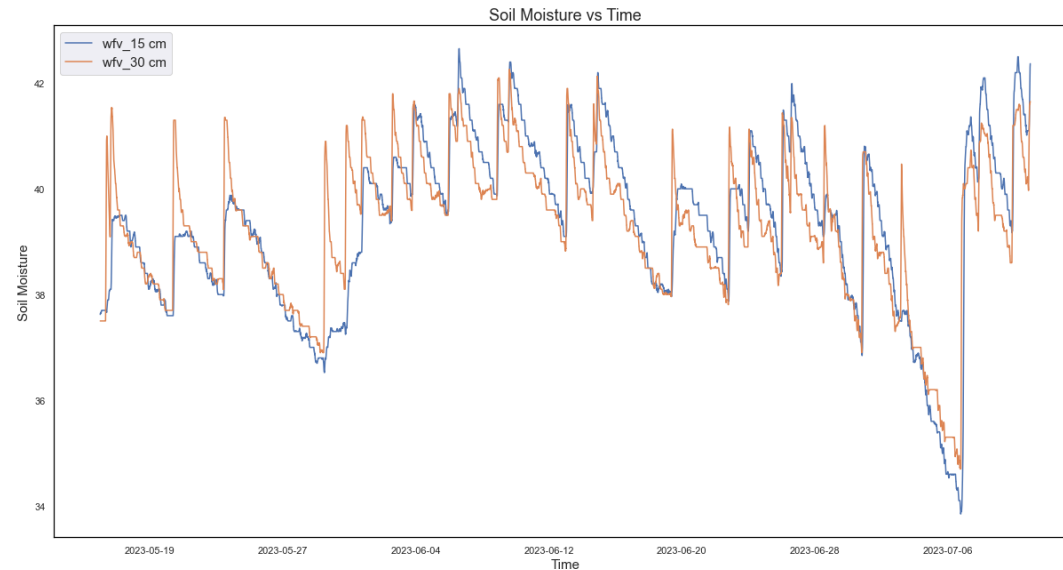


# Open-source ET data for half of the growing season



- Hourly ET data for the same time frame was obtained from Open-Meteo.com API using The United States National Oceanographic and Atmospheric Administration Global Forecast System (NOAA GFS)

# Available Data Set



- Time Series data at two depths for about half the growing season
- The dewatering time series data
- Below field capacity between irrigation cycles where

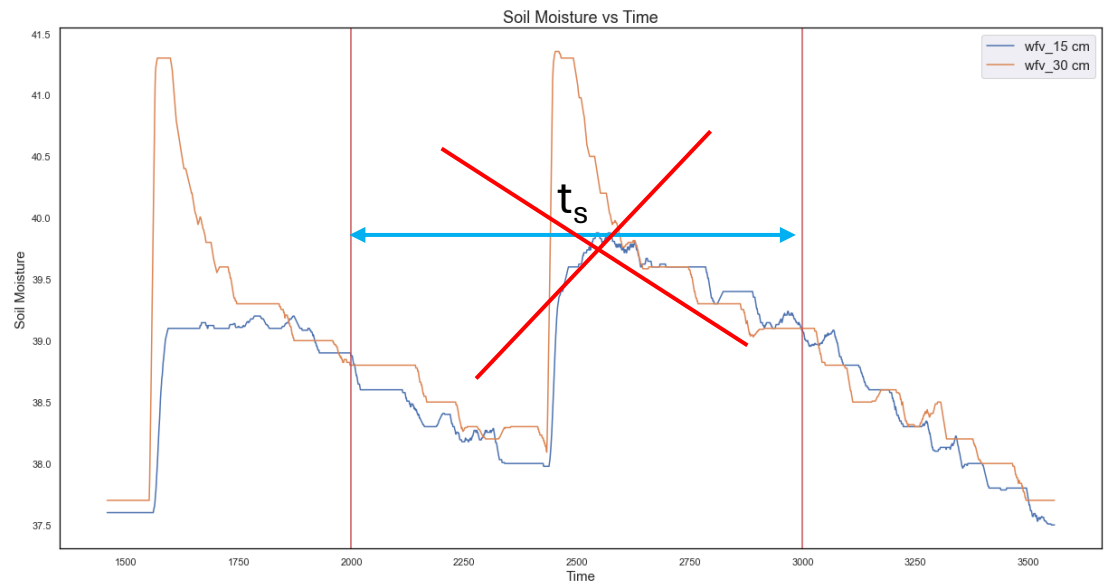
$$\frac{\partial \theta}{\partial t} = \text{dewatering flux}$$

$R^2$  is 0.98 during ET driven dewatering.

# Aggregating Soil Moisture Training Data



| Hours – $t_s$ values | Number of occurrences in growing season |
|----------------------|-----------------------------------------|
| 30                   | 163                                     |
| 40                   | 163                                     |
| 48                   | 163                                     |
| 60                   | 163                                     |
| 65                   | 168                                     |
| 72                   | 155                                     |
| 80                   | 162                                     |
| 96                   | 146                                     |
| 108                  | 47                                      |
| 120                  | 42                                      |
| 144                  | 289                                     |



| Target              | Input Variables |                        |             |
|---------------------|-----------------|------------------------|-------------|
| $Y^{(i)}$           | $X_1^{(i)}$     | $X_2^{(i)}$            | $X_3^{(i)}$ |
| $\theta_{ts}^{(1)}$ | $\theta^{(1)}$  | $\sum_i^{ts} ET^{(1)}$ | $t_s^{(1)}$ |
| $\theta_{ts}^{(2)}$ | $\theta^{(2)}$  | $\sum_i^{ts} ET^{(2)}$ | $t_s^{(2)}$ |
| $\theta_{ts}^{(3)}$ | $\theta^{(3)}$  | $\sum_i^{ts} ET^{(3)}$ | $t_s^{(3)}$ |

# Incorporating soil properties into the ML Model

$$\theta = \frac{\theta_s - \theta_r}{[1 + |\alpha h|^n]^m} + \theta_r$$

[1]

By rearranging the van Genuchten equation, we can use the soil survey parameters to estimate unsaturated hydraulic head and unsaturated hydraulic conductivity

$$h = \frac{\left( \left[ \frac{\theta_s - \theta_r}{\theta - \theta_r} \right]^{1/m} - 1 \right)^{1/n}}{\alpha}$$

[2]

| Parameter | $\theta_s$ (wfv) | $\theta_r$ (wfv) | $\alpha$ | n    | $K_s$ (cm/s) | $l\ell$ |
|-----------|------------------|------------------|----------|------|--------------|---------|
| Silt Loam | 0.439            | 0.065            | 0.005058 | 1.66 | 1.75         | 2.317   |

# Incorporate soil survey data in the ML Model

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left( K \left( \frac{\partial h}{\partial z} \right) \right) + \frac{\partial K}{\partial z} - S \quad [3]$$

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left( K \left( \frac{\partial h}{\partial z} \right) \right) - ET \quad [4]$$

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad [5]$$

$$K = K_s S_e^\ell \left[ 1 - \left( 1 - S_e^{1/m} \right)^m \right]^2 \quad [6]$$

$$\frac{\partial}{\partial z} \left( K \left( \frac{\partial h}{\partial z} \right) \right) = W^{(t2)} = K_{1+1/2}^{(t2)} \left( \frac{h_1^{(t2)} - h_2^{(t2)}}{\Delta z^2} \right) - K_{2+1/2}^{(t1)} \left( \frac{h_2^{(t2)} - h_3^{(t2)}}{\Delta z^2} \right) + \frac{K_{1+1/2}^{t2} - K_{2+1/2}^{(t2)}}{\Delta z} \quad [7]$$

- We Start with the 1-D Richards Equation
- The Darcian part of the Richards equation can be removed.
- The unsaturated hydraulic head can be calculated from the effective saturation.
- With numerical methods (first published by Celia 1990).

# Machine Learning Model Setup

$$X \in \mathbb{R}^{1260 \times 12}$$

$$\begin{bmatrix} x_1^{(1)} & \dots & x_{12}^{(1)} \\ \vdots & \ddots & \vdots \\ x_1^{(1260)} & \dots & x_{12}^{(1260)} \end{bmatrix}$$

$$Y \in \mathbb{R}^{1260 \times 1}$$

| Input variable feature                                                 | Symbol                      |
|------------------------------------------------------------------------|-----------------------------|
| Soil Moisture at time $t^{(i)}$                                        | $\theta^{(i)}$              |
| Timestep in filtered subset corresponding to the row of input features | $t^{(i)}$                   |
| Time from time $t^{(i)}$ to the target predicted soil moisture.        | $t_s^{(i)}$                 |
| Rolling slope prior 12hours from time $t^{(i)}$                        | $r_{\text{slope12}}^{(i)}$  |
| The sum hourly ET Prior 24 hours from time $t^{(i)}$                   | $ET_{\text{prior24}}^{(i)}$ |
| The sum of the hourly ET between time $t^{(i)}$ and time $t_s^{(i)}$   | $ET_{\text{post}}^{(i)}$    |
| Hydraulic Head                                                         | $h^{(i)}$                   |
| Unsaturated Hydraulic conductivity                                     | $K^{(i)}$                   |
| Simplified capillary coefficient                                       | $W_t^{(i)}$                 |
| Sum of the hourly ET 24 hours after $t^{(i)}$                          | $ET_{\text{post24}}^{(i)}$  |

## Skikit-learn Regressors:

- Support vector Machine
- Ridge Regression
- Random Forest
- Lasso Regression

## Regularization:

- Mean Square Error (MSE) plus other regularizations were used for gradient descent
- Grid and Random Searches were used for optimization.

## Validation and Training

- Validation data set
- 15% Holdout from the training



# Results and Discussion

|       | Random Forest | Support Vector Machine | Lasso Regression | Ridge Regression |
|-------|---------------|------------------------|------------------|------------------|
| $r^2$ | 0.998         | 0.947                  | 0.980            | 0.980            |
| RMSE  | 0.13          | 0.87                   | 0.50             | 0.50             |

**Results of the 15% holdout from the 1260-member training set. Using all the input features.**

|       | Random Forest | Support Vector Machine | Lasso Regression | Ridge Regression |
|-------|---------------|------------------------|------------------|------------------|
| $r^2$ | 0.62          | *                      | 0.83             | 0.905            |
| RMSE  | 3.02          | 4.53                   | 1.81             | 1.36             |

**Results on the validation set using all of the input features.**

|       | Random Forest | Ridge Regression |
|-------|---------------|------------------|
| $r^2$ | 0.63          | 0.93             |
| RMSE  | 3.04          | 1.13             |

**Validation data set with  $h^{(i)}$ ,  $K^{(i)}$ ,  $W_t^{(i)}$  removed from the training data set.**

## Discussion:

- Ridge Regression performed reasonably well on the validation data set, considering the ET data was generated from a weather station 8 km away.
- Random Forest performed well on the holdout but performed poorly on the validation data set because the validation data set includes data that is outside the range of the training.
- The capillary parameters calculated from soil moisture values and the soil survey did not add much value, however, monitoring more depths and a site-specific soil survey could improve the model.

# Conclusion

- A model was made using the dewatering portion of time series soil moisture data, ET during the growing season and soil survey information to predict future soil moisture values.
- Four Machine Learning regressors were evaluated.
- Conclusions:
  - Soil Survey information did not improve the model although we suspect that monitoring more than two depths and using a site-specific soil survey could offer improvements.
  - Ridge regression had reasonable performance on a validation data set with an  **$R^2 = 0.93$**  and a **RMSE = +/- 0.011 wfv** even though the ET data was 8 km away. Site specific ET data could also be a source of improvement.
  - This model serves as a proof of concept that time series soil moisture and ET data, basic information accessible by growers, can be aggregated into a ML regression for prediction of future soil moisture values.



# References

- Celia, M. A. (1990). A general mass- conservative numerical solution for the unsaturated flow equation. *Water Resources Research*, 26(7), 1483-1496.
- Cortes, C. V. (1995). Support-Vector Networks. *Machine Learning*. 20, 273-297.
- Ghorbani, A. M. (2021). Towards new soil water flow equations using physics-constrained machine learning. *Vadose Zone J.* , 20:e20136.
- Ireson, A. M. (2023). A simple, efficient, mass-conservative approach to solving Richards' equation. *Geosci. Model Dev.*, 16, 659-677.
- Lianga, X. V. (2016). Scheduling irrigation using an approach based on the van Genuchten model. *Agricultural Water Management* 176 , 170-179.
- Pachepsky, Y. ,. (2003). Generalized Richards' equation to simulate water transport in unsaturated soils. *Journal of Hydrology* 272, 3-13.
- Richards, L. (1931). Capillary conduction of liquids in porous mediums. *Physics*, 1, 318-333.
- Seyfried, M. L., & Seyfried. (2005). Dielectric loss and calibration of the Hydra Probe soil water sensor. *Vadose Zone J.* 4:, 1070-1079.
- Suleiman, A. A. (2003). Modeling Soil Water Redistribution during Second-Stage Evaporation. *Soil Sci. Soc. Am. J.* 67, 377-386.
- van Genuchten, M. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Sci. Soc. Am.*, 44, 892-898.
- Warrick, A. W. (2003). *Soil Water Dynamics*. Oxford University Press.
- Zarba, R. L. (1988). *A Numerical Investigation of Unsaturated Flow*. Massachusetts Institute of Technology.

# Thank You!

And thanks to the Cascade Berries Farms  
Sam Kuschnick



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